

SolarChain: A Physics-Grounded Embodied IoT System for Verifiable Urban Solar Market Design

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Code, data, and replay artifacts
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50
simulated PV nodes
5 Chinese cities

36,000
hourly records
720 steps; fixed seeds

11
scripted attack classes
7 detector variants

144,721
hash-linked events
all links reproduced

Central result. SolarChain links **physics-bounded sensing evidence** to planner-mediated allocation and contract settlement, while separating **replayable consistency** from **unproven source truth**.

Research problem and contribution

Distributed solar markets connect physical reports to digital incentives. A ledger preserves a submitted value, but cannot determine whether the underlying generation report was physically plausible.

Research question

Can a controlled EIoT prototype carry physics-bounded PV reports through coordinated settlement while retaining replayable decision evidence?

- 1 **System**: screening, persistent agents, planner approval, configurable allocation, and contract settlement.
- 2 **Evaluation**: detector accuracy, 11 attacks, allocation sensitivity, runtime, and audit integrity.
- 3 **Artifacts**: an open benchmark separating sourced weather, modeled physics, and synthetic behavior.

EIoT realization and physical evidence

Enacted City weather, PV geometry, and physical constraints bound each report.

Engaged Solar, Planner, Factory, and Market roles coordinate verification and settlement.

Evolutionary Residual, calibration, trust, and participation state persist; adaptation is rule-based.

area A , efficiency η , irradiance G , coefficient β , and reference temperature T_{ref} :

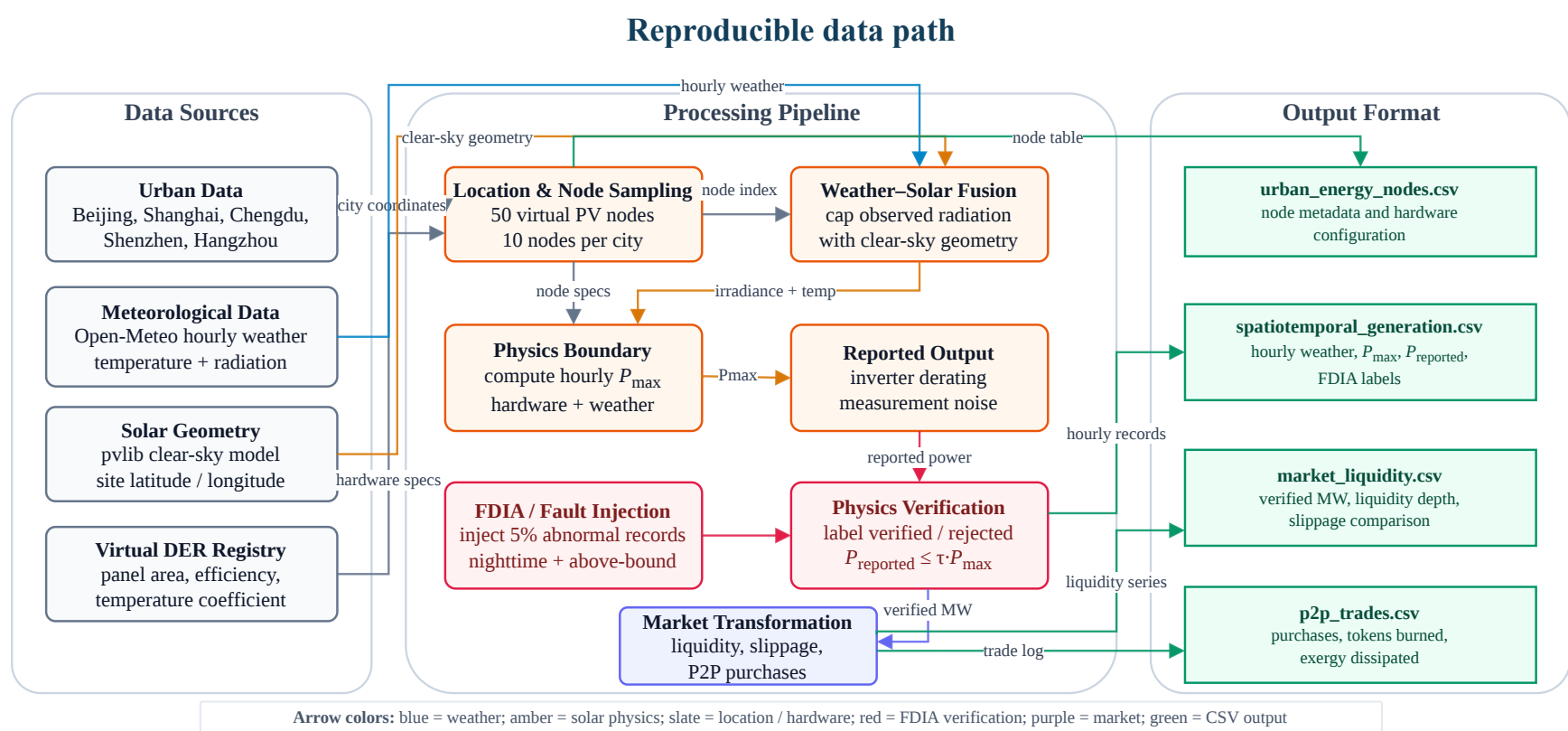
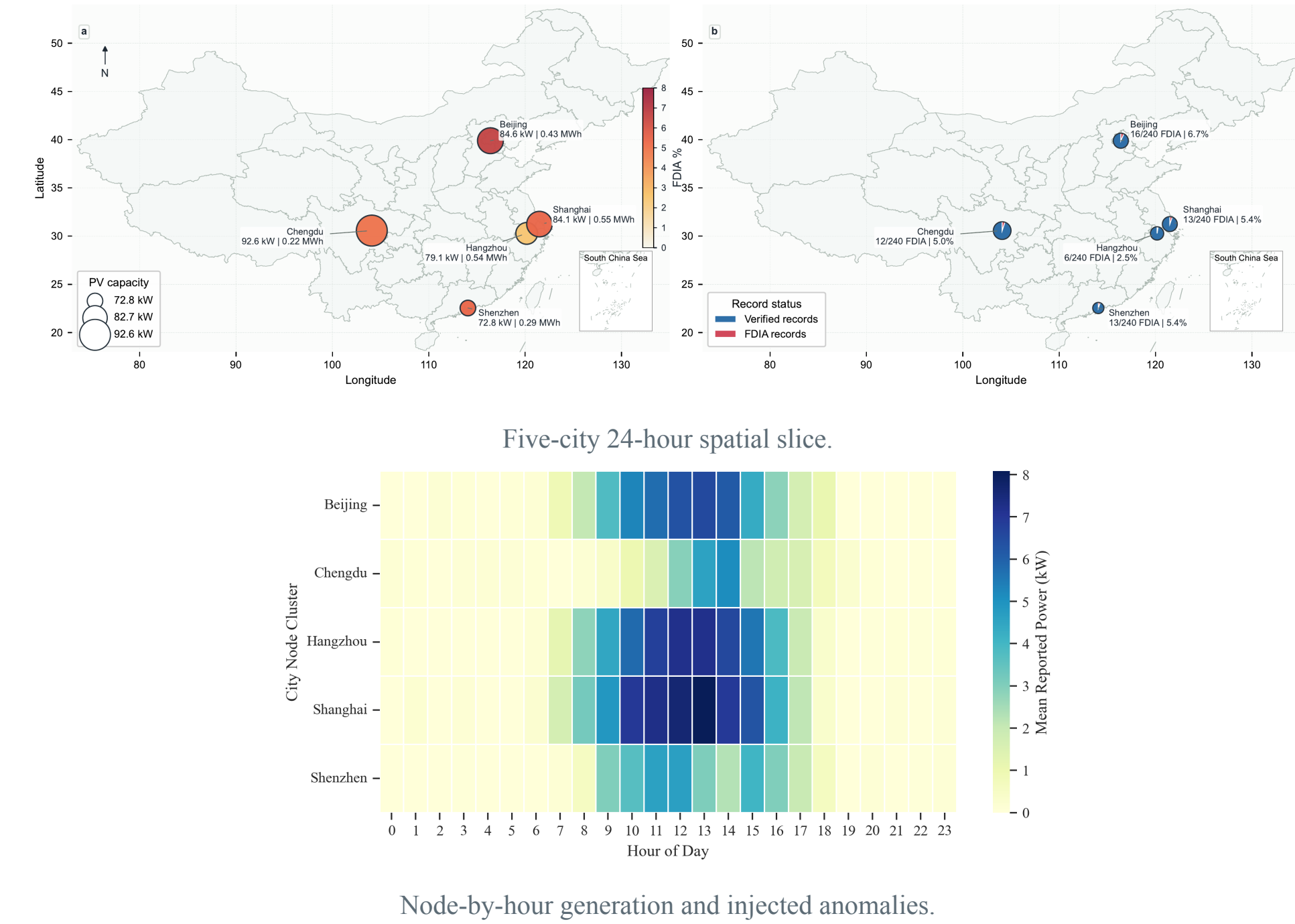
$$P_{\max} = A\eta G[1 - \beta(T - T_{ref})], \quad P_{rep} > \tau P_{\max} \Rightarrow \text{reject.}$$

Temporal, neighborhood, and residual-memory rules add context.

Trust boundary **Physical plausibility is testable.** **Source authenticity is not**: spoofed weather, coordinated near-bound reports, or compromised sensors can remain plausible.

Controlled benchmark and provenance

Window 1 Apr–1 May 2026; 720 hourly steps
Scale 50 nodes; 5 cities; 36,000 records
Labels 1,800 original FDIA records (5%)
Sourced Open-Meteo temperature and shortwave radiation
Modeled Solar position, clear-sky envelope, correction, $P_{\max}$
Synthetic Hardware, reports, demand, trades, agents, attacks
Protocol Fixed seeds, cached weather, file hashes, released scripts



Inputs $\rightarrow$ physics $\rightarrow$ FDIA verification $\rightarrow$ market outputs.

Reproduce and inspect

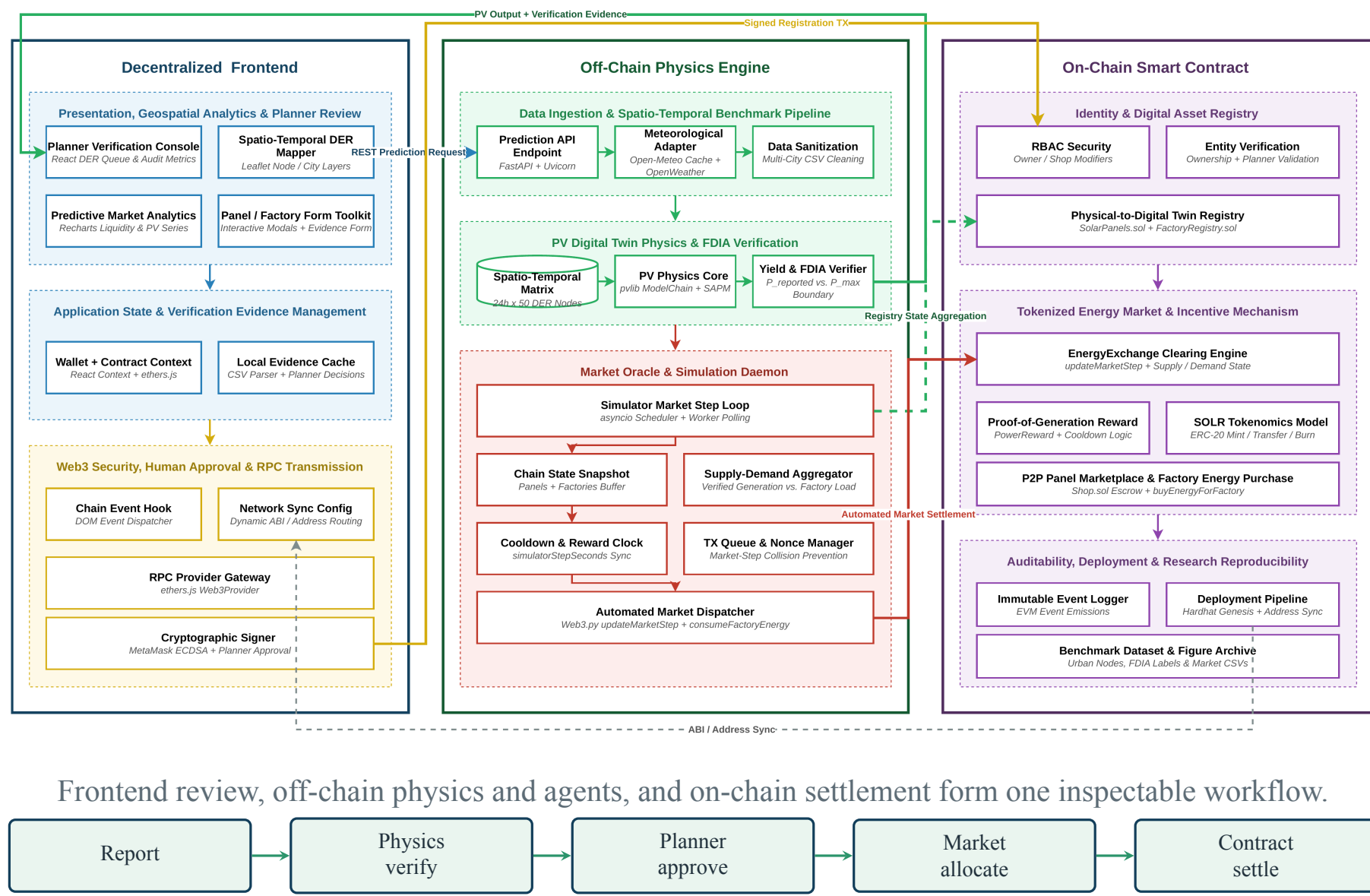
python Simulator/data/generate_monthly_datasets.py
python Simulator/experiments/run_all_eiot_experiments.py
python Simulator/data/visualizations.py **Data availability**: configurations, plots, tests, and replay scripts are linked by the QR code. Weather is cached; market behavior is simulated.

Provenance taxonomy

- Sourced** Open-Meteo weather and released city coordinates
- Modeled** Solar geometry, temperature response, and $P_{\max}$
- Synthetic** Virtual assets, reports, attacks, demand, and trades
- Replayed** Detector decisions, market rows, purchases, and event hashes

Integrity check: cached inputs and serialized configuration prevent silent data drift between benchmark regeneration and result replay.

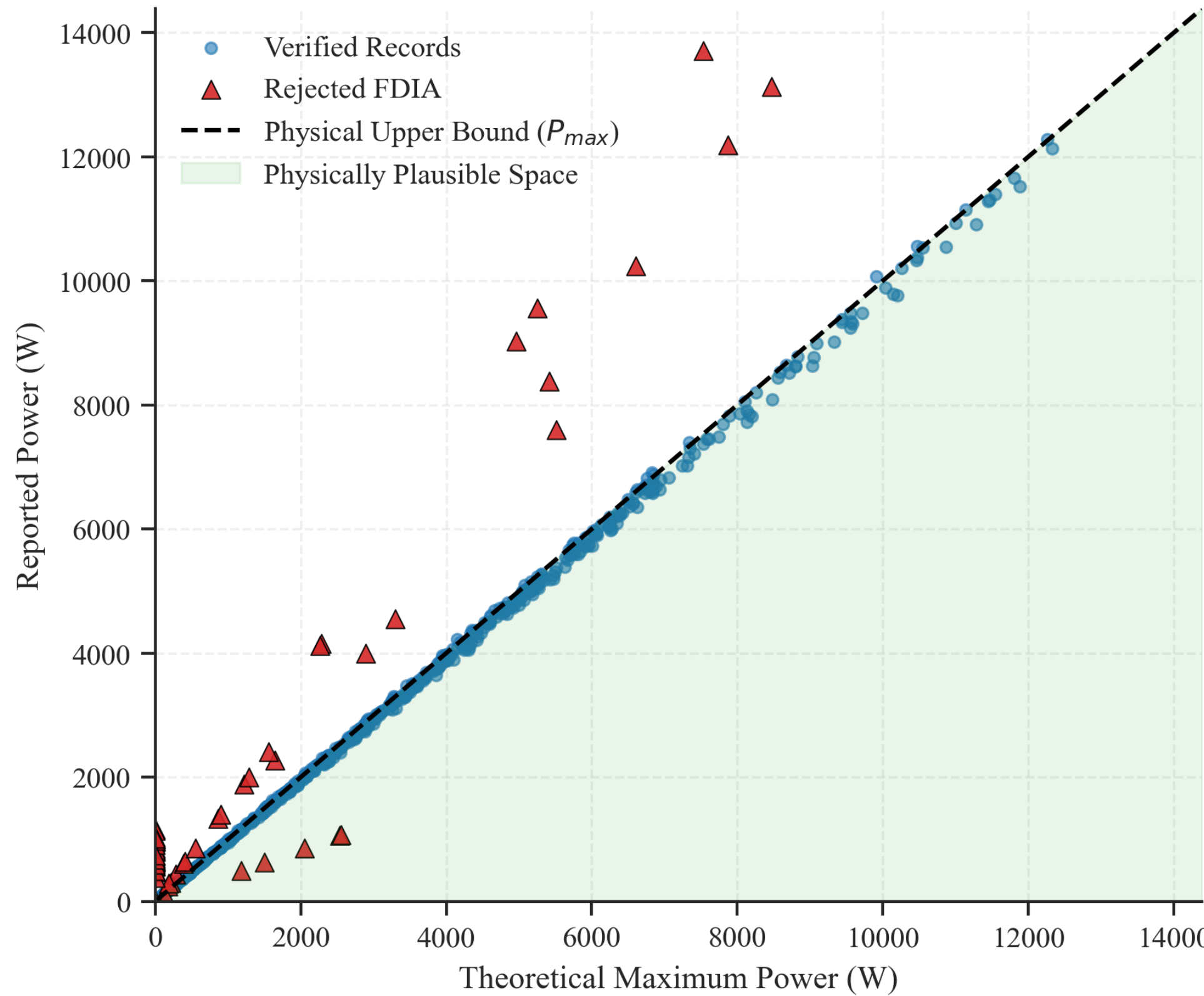
End-to-end system architecture



Frontend review, off-chain physics and agents, and on-chain settlement form one inspectable workflow.

Verification results

Detector	Prec.	Rec.	F1	FAR	FRR	$\mu\text{s}/\text{rec.}$
3-sigma	1.000	.766	.868	.234	.000	.177
IQR/MAD	1.000	1.000	1.000	.000	.000	.218
Physics bound	1.000	.862	.926	.138	.000	.004
Adaptive agent	.976	1.000	.988	.000	.001	1.449



Reported power versus theoretical maximum; red triangles are injected attacks.

- 1 **IQR/MAD** is best for aggregate classification on this injected distribution.
- 2 The **adaptive agent** is slightly less accurate, but retains residuals, temporal context, and state transitions.
- 3 Explanation does not compensate for attack classes outside the evidence model.

Attack-class boundary and runtime

Capability	Attack classes	F1
Strong	Original; nighttime; above-bound; neighbor-inconsistent	.988–1.000
Partial	Sensor drift; replay; intermittent burst	.788–.941
Outside	Near-bound; within-bound; weather spoof; coordinated near-bound	.000

time: 30 warmed, full-batch runs. Median $\mu\text{s}/\text{record}$: physics 0.004; 3-sigma 0.177; IQR/MAD 0.218; adaptive agent 1.449.

How verification error reaches the market

False accept	Invalid energy enters reward, liquidity, and settlement.
False reject	Valid supply is withheld, increasing unmet demand.
Context error	Spoofed exogenous data shifts the physical envelope.
Interpretation	Detector choice is an economic control, not only a classification decision.

$$E_t^{acc} = \sum_i a_{i,t} E_{i,t} \quad \Delta R_t = \rho \Delta E_t^{acc} \quad \Delta L_t = (1 - \rho) \Delta E_t^{acc}$$

Decision evidence retained per report

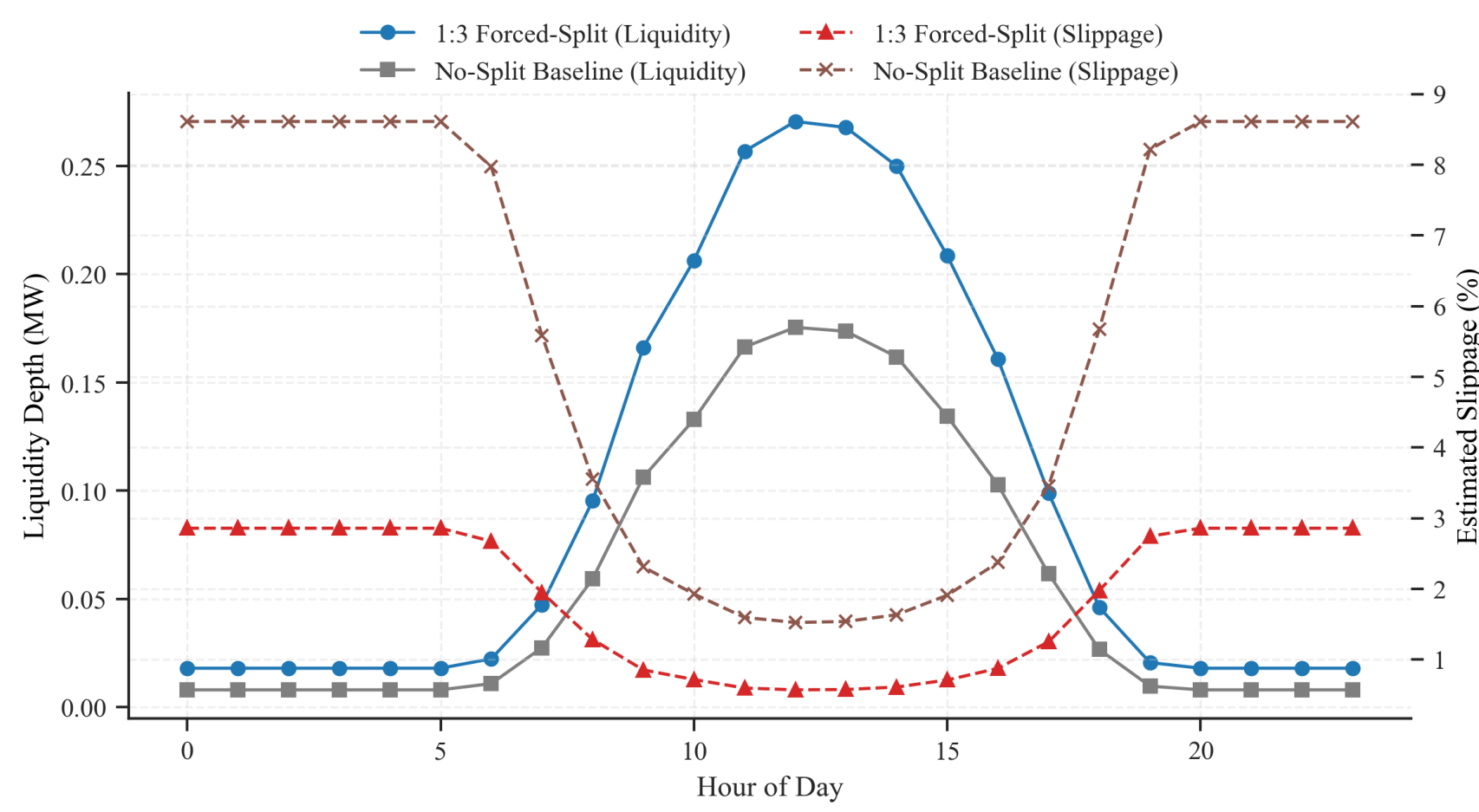
- Physical** $P_{\max}$, exceedance margin, and nighttime consistency
- Statistical** Robust deviation score and neighborhood comparison
- Temporal** Recent residual history, calibration, and trust state

Configurable reward-liquidity allocation

Accepted one-hour energy is split by reward share ρ :

$$R_t = \rho E_t^{acc}, \quad L_t = (1 - \rho) E_t^{acc}.$$

The 41-point sweep evaluates seven scenarios. Under declared weights and feasibility constraints, the selected default is **20% producer reward / 80% market liquidity**.

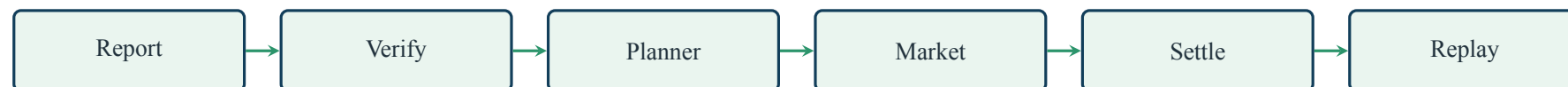


Greater modeled liquidity depth reduces the slippage proxy.

Reward/liquidity	Fulfill	Reward	Slip.	Unmet	Score
0/100	.961	0	32.36%	1.869 MWh	.617
10/90	.964	4,174	32.17%	1.686 MWh	.640
20/80	.956	8,976	32.57%	2.111 MWh	.651
50/50	.782	22,441	33.30%	9.520 MWh	.592

shares span 12.5–22.5%. The 20% result is conditional on the simulator, scenarios, normalization, constraints, and weights.

Replayable audit chain



Each event names the previous hash:

$$h_t = \text{SHA256}(h_{t-1} \| e_t \| d_t \| m_t).$$

All **144,721** events reproduced their stored linkage. Local-EVM tests cover registration, access control, ratio bounds, SOLR burning, and settlement fields. Rejected generation is excluded downstream.

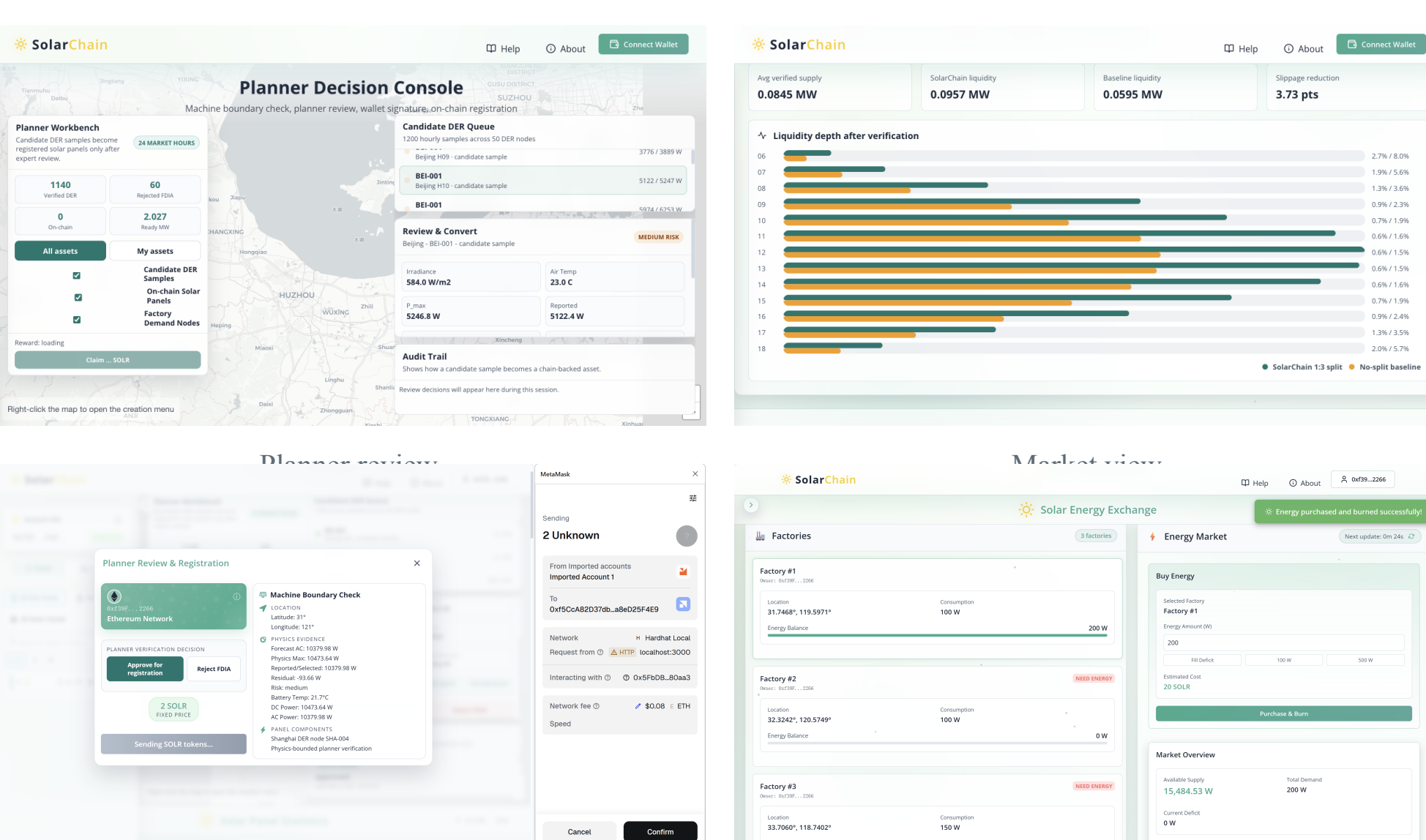
Verify Reward plus liquidity equals accepted energy.

Allocate Hashes, city totals, and purchases satisfy released invariants.

Replay

What replay proves
The recorded path has not changed. **It does not prove** source truth, calibration, asset control, or contract security.

Working prototype



Verified supply.

Factory purchase.

Accepted supply 2,028.13 kWh across five synthetic city clusters

Purchase trace 42 trades; 0.6052 MWh; 609.77 SOLR-equivalent units burned

Meaning Software linkage, not real transactions or carbon abatement

Evidence boundary and next validation

Evidence class	What is evidenced	Status
Verification	36k labels, 11 attacks, physics bounds, 144,721-event replay	Supported
Market	41 allocation points across 7 scenarios	Simulated
Deployment	No trusted meter or governed field pilot	Open

Required before field claims: synchronized inverter and meter traces; seasonal faults and adaptive attacks; privacy, latency, loss, congestion; ratio governance, appeals, and a governed pilot.

Take-home
A strong detector may lack source evidence; a favorable market score may depend on accepted false supply; perfect replay may preserve a false input.

System synthesis From physical report to bounded claim

STAGE 1
Sense
 $P_{i,t}^{cp}$
Weather and PV specifications define context.

STAGE 2
Verify
 $a_{i,t} \in \{0, 1\}$
Physics and temporal context gate each report.

VERIFICATION
Invariant: exclude rejected energy.
Risk: false accept or bypass.
Owner: verifier / planner

ALLOCATION
Invariant: $R_t + L_t = E_t^{acc}$.
Risk: hidden or unmatched split.
Owner: market governance

REPLAY
Invariant: bind prior hash and payload.
Risk: broken recomputation.
Owner: contract / audit operator